

WEEK PRACTICAL – 01

```
saicharan_reddi@SAICHARANREDDY:~# id
uid=0(saicharan_reddi) gid=0(saicharan_reddi) groups=0(saicharan_reddi)
```

1.id:

Report: After running the id command, I could see my user details. It showed my username as **Charan**, along with the user ID, group ID, and the groups I belong to like sudo and plugdev.

2.env:

```
_DISTRO_NAME=Ubuntu
ME=SAICHARANREDDY
D=/root
GNAME=root
TD_SHOWN=update-motd
ME=/root
NG=C.UTF-8
_INTEROPP=/run/WSL/392_interop
_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33:0
:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lh
t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01
:l=31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01
01;31:*.7z=01;31:*.xz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;
if=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm
:mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.
35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.
*:xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgy=01;35:*.emf=01;35:*.ogv=01;35:*.
*:mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.
download=00;90:*.dpkg-dist=00;90:*.dpkg-new=00;90:*.dpkg-old=00;90:*.dpkg-
ig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-ne
YLAND_DISPLAY=wayland-0
SSCLOSE=/usr/bin/lesspipe %s %s
RM=xterm-256color
SSOPEN=| /usr/bin/lesspipe %s
ER=root
SPLAY=:0
LVL=1
G_RUNTIME_DIR=/run/user/0
LENV=
G_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
TH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games
avapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt
/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/windows
t/c/Program Files/nodejs:/mnt/c/Users/saicharan_6wg39hp/AppData/Local/Microso
/bin:/mnt/c/Program Files/JetBrains/PyCharm 2026.1/bin:/mnt/c/Users/saichar
_reddi/AppData/Local/Office/12.1.0.274586/clitool:/snap/bin
US_SESSION_BUS_ADDRESS=unix:path=/run/user/0/bus
STTYPE=x86_64
LSE_SERVER=unix:/mnt/wslg/PulseServer
/usr/bin/printenv
saicharan_reddi@SAICHARANREDDY:~#
```

Report: When I executed the env command, many environment variables were displayed. I observed variables like HOME=/home/charan, USER=charan, PWD=/home/charan/OSSP, and PATH, which contain information about the system environment.

3.Export:

```
35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.~=00;90:*.#=00;90:*.bak=00;90:*.crdownload=00;90:*.dpg=00;90:*.dpg-dist=00;90:*.dpg-new=00;90:*.dpg-old=00;90:*.dpg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:"
declare -x NAME="SRIVALLI"
declare -x OLDPWD
declare -x PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Users/sriva_6wg39hp/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/sriva_6wg39hp/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Program Files/JetBrains/PyCharm 2026.1/bin:/mnt/c/Users/sriva_6wg39hp/AppData/Roaming/npm:/mnt/c/Users/sriva_6wg39hp/AppData/Local/Kingsoft/WPS Office/12.1.0.27458/clitool:/snap/bin"
declare -x PULSE_SERVER="unix:/mnt/wslg/PulseServer"
declare -x PWD="/root"
declare -x SHELL="/bin/bash"
declare -x SHLVL="1"
declare -x TERM="xterm-256color"
declare -x USER="root"
declare -x WAYLAND_DISPLAY="wayland-0"
declare -x WSL2_GUI_APPS_ENABLED="1"
declare -x WSLENV=""
declare -x WSL_DISTRO_NAME="Ubuntu"
declare -x WSL_INTEROP="/run/WSL/643_interop"
declare -x XDG_DATA_DIRS="/usr/local/share:/usr/share:/var/lib/snapd/desktop"
declare -x XDG_RUNTIME_DIR="/run/user/0"
```

Report: The export command displayed the exported environment variables. I observed that variables such as HOME, USER, PWD, PATH, and other system variables were available and exported for the current session

4.Alias:

```
saicharan_reddi@SAICHARANREDDY:~$ alias
alias
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
alias ls='ls --color=auto'
```

Report: After executing the alias command, the terminal displayed a list of shortcut commands. I observed aliases like grep, ls, and ll, which are used as shortcuts for commonly used commands

WEEK-2 Practical

TASK-1 Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to 3 another. Explain how control transitions between user Space and kernel space during execution

```
saicharan_reddi@SAICHARAN x + v
GNU nano 8.7.1 copyfile.c *
#include <unistd.h>
#include <stdlib.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[])
{
    int source, destination;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3)
    {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    source = open(argv[1], O_RDONLY);

    if (source < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    destination = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    while ((bytes_read = read(source, buffer, BUFFER_SIZE)) > 0)
```

```
saicharan_reddi@SAICHARANREDDY:~$ nano filecopy.c
saicharan_reddi@SAICHARANREDDY:~$ gcc -Wall -Wextra filecopy.c -o bin/filecopy
saicharan_reddi@SAICHARANREDDY:~$ echo "This is a sample file for system call testing." > sample.txt
saicharan_reddi@SAICHARANREDDY:~$ ./bin/filecopy sample.txt copy.txt
File copied successfully.

saicharan_reddi@SAICHARANREDDY:~$ cat sample.txt
This is a sample file for system call testing.

saicharan_reddi@SAICHARANREDDY:~$ cat copy.txt
This is a sample file for system call testing.
```

Report: I developed a C program to copy the contents of one file to another using the system calls `open()`, `read()`, `write()`, and `close()`. The source file was opened in read-only mode and the destination file was created for writing. The contents were read into a buffer and written to the

Task-2 : Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system 3 calls and identify the kernel services involved

Report: I used the strace utility to trace the system calls generated by the command `cat sample.txt`. The trace showed important system calls such as `openat()`, `read()`, `write()`, and `close()`. `openat()` opens the file, `read()` obtains the file contents, `write()` displays the contents on the terminal, and `close()` closes the file descriptor. The experiment helped me understand how a user-level command interacts with kernel services through system calls.